

Cleaning a Corrupted Temperature Anomaly Dataset

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I. INTRODUCTION

The file `global_temp_dirty_v2.csv` holds corrupted monthly global temperature anomalies ($^{\circ}\text{C}$, relative to the 1901–2000 mean) for January 1880 to December 2025. This report shows the counts produced by the code `clean.py`, the normalization applied, and the five warmest years. Fig. 1 shows the cleaned series.

II. DATA CLEANING SUMMARY

Load and standardize dates All fields were read as text and header names stripped. Of 1711 lines after the header, 4 non-data lines were discarded, 32 rows with swapped columns were fixed. Ten date patterns were matched by explicit regular expressions (two-digit years: 26–99→19xx, 00–25→20xx).

Parse values 34 comma-decimal and 45°C -suffixed values were converted to `float`; 63 missing-value tokens became `NaN`.

Deduplicate and sort After chronological sorting, 20 duplicate rows were removed, leaving 1687 unique months.

Remove outliers with the IQR method Of the 1625 non-missing values, $Q_1 = -0.385$, $Q_3 = 0.49$, $\text{IQR} = 0.875$, so the fences are $[-1.6975, 1.8025]$ $^{\circ}\text{C}$. Exactly 63 values fell outside: and they were set to `NaN`.

Impute missing months On the full 1752 month grid the 65 absent months, the missing tokens and the removed outliers gave 190 gaps, all filled by linear interpolation in time.

III. NORMALIZATION

From the cleaned monthly series x two quantities were computed:

$$d = x - \mu_{20}, \quad (1)$$

$$z = (x - \mu)/\sigma, \quad (2)$$

where x is the cleaned monthly anomaly ($^{\circ}\text{C}$), d is its departure from the 20th-century mean μ_{20} (the mean of the cleaned series over 1901–2000), and z is its standardized score, μ and σ being the mean and sample standard deviation over 1880–2025. Here $\mu_{20} = 0.0002^{\circ}\text{C}$ (near zero, as expected for a 1901–2000 baseline), $\mu = 0.464^{\circ}\text{C}$ and $\sigma = 0.5199^{\circ}\text{C}$.

IV. WARMEST YEARS

Calendar-year means of x and z give the five warmest years in Table I. 2024 and 2023 lie more than 2σ above the long-run mean.

Fig. 1 draws the series as a `LineCollection` whose segments are colored by d : blue below the 20th-century mean, red above.

TABLE I
FIVE WARMEST CALENDAR YEARS IN THE CLEANED SERIES.

Rank	Year	Mean anomaly ($^{\circ}\text{C}$)	Mean z
1	2024	+1.168	+2.157
2	2023	+1.153	+2.129
3	2025	+1.013	+1.858
4	2016	+0.982	+1.799
5	2022	+0.960	+1.758

Note: AI was used to evaluate and improve the regex patterns required for parsing various date formats and values. It was also used to verify compliance with the required IEEE latex format.

`pdflatex report.tex` command was tested and verified but there is noted limitation that the command needs to be run twice to produce the numbering of the figures and tables referenced by the text.

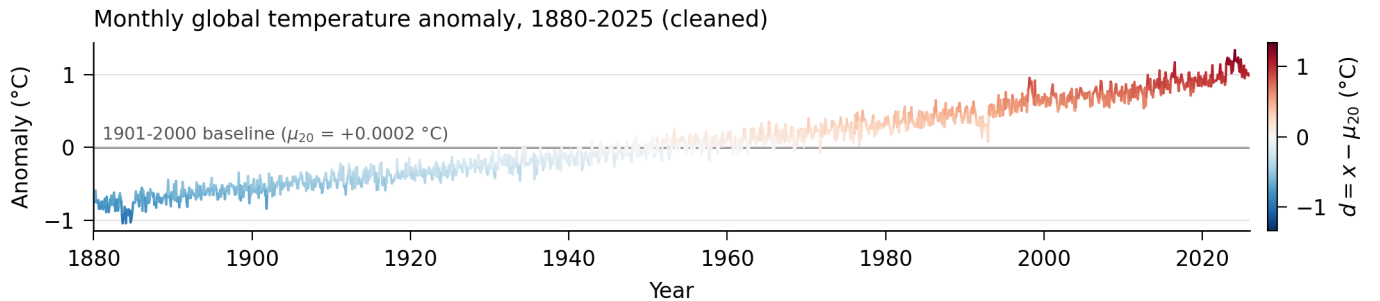


Fig. 1. Cleaned monthly global temperature anomaly, 1880–2025. Position encodes the anomaly in $^{\circ}\text{C}$; segment color encodes $d = x - \mu_{20}$ on a diverging scale centered at the 1901–2000 mean.